S3 Supplementary QC and Robustness

Export validity, missingness, and analysis boundaries

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Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

S3.1 Export QC

The run manifest reports 70 H5 files with 3D MNPS, 9D MNPS, 3D MNJ, and 9D MNJ in every session. This is the artifact-masked re-export (`..._20260805_132612`); the ingest pipeline imported 715 publicly released BAD_-interval artifact annotations (2,143.7 s total) and masks flagged samples before epoching, dropping any 8-s epoch that overlaps a masked interval rather than bridging across it. The visit-level export confirmed `ok` geometry status for 69/70 sessions and `adjusted` status for 1/70 (down from 68/2 in the pre-masking export); all are accepted under the standard invalidity policy. 9D Jacobians resolve in 70/70 sessions. The median epoch count per session was 78.5 (range 20–109), down from a pre-masking median of 98 (range 30–148), and the total usable epoch pool fell from 6,929 to 5,265. TD25v4 has no BAD_ annotation but was already coverage-excluded from this export, so its absence does not affect any analysed session. These checks validate availability under the export contract; they do not validate a developmental biological interpretation.

S3.2 Coordinate-contract boundary

The export contains subject- and cohort-anchored 3D/9D layers. The cohort layers use the frozen subject-balanced global IQR anchor `figshare_infant_eeg_global_cohort_all_visits_iqr_v2_1`, refit on the artifact-masked epoch pool (anchor hash `45bbfa21...c301a89e`); this enables common-chart coordinate modelling. It is not an independent normative or held-out-age reference. Local flow/reachability and coordinate positions remain separate analyses.

S3.3 Longitudinal missingness

Twenty-two children contributed 70 visits. Twenty had at least two visits and could in principle contribute to within-child estimation; two had only a single visit and cannot contribute to any within-child family. Recording ages ranged from 38 to 203 days and visit labels are ordinal rather than age strata. Within those 20 multi-visit children, eligibility still varies by family and endpoint completeness: the primary 3D within-child family (R4) retains 20 children/68 endpoint-complete visits, while the direct-age-only sensitivity subsets (S2.4, S2.7, S2.8) retain only 18 children/36 visits after also requiring a spreadsheet-direct age value, so 2-4 further multi-visit children drop out depending on which family and age-source restriction is applied. This design cannot support a balanced 4-, 8-, or 12-month paired analysis.

S3.4 Sensitivity summary

The primary cohort-coordinate GEE family is FDR-supported in 12/12 endpoints, and the direct-age sensitivity supports 11/12 (the 3D entropy-related coordinate is the exception). The within-child MNJ/reachability robustness family (S2.5-S2.7) and the cohort-pooled reachability robustness family (S2.8) are not supported after FDR sensitivity in any of their tested forms. Their primary forms are not nominally significant, but the cohort-pooled direct-age sensitivity retains one nominal hit: the family-wise minima are raw p=0.054 (3D within-child, MNJ trace), raw p=0.076 (9D MNJ sensitivity), raw p=0.208 (3D within-child direct-age subset), raw p=0.196 (cohort-pooled reachability primary), and raw p=0.040 (cohort-pooled reachability direct-age subset, not FDR-supported). Two 3D within-child endpoints that were nominally significant in the direct-age subset before artifact masking (raw p=0.043-0.050) are no longer nominal (raw p=0.208-0.277) after masking; this is reported descriptively in S2.7, not as a formal ablation test. No event-locked, stage-stratified, artifact-balance, or Bayley-outcome analysis is claimed because the relevant harmonized inputs were not available for a defensible primary test.

S3.5 Within/between decomposition and cluster-permutation robustness (M3, M4)

The within/between decomposition (S2.2) shows the primary cohort-coordinate result is attributable to within-child change at all 12 endpoints (within-child raw $p < 0.011$ throughout), not to between-child differences (no

endpoint's between-child term is FDR-significant). However, the child-cluster row-permutation check (S2.3) – run precisely because 22 clusters is a known regime for anti-conservative GEE sandwich-covariance inference – supports only 6 of 12 endpoints after its own FDR correction (9D m_o, 3D deviation, 9D m_e, 9D d_n, 9D e_e, 9D e_m). The 3D mobility and 3D entropy-related coordinates are the least robust endpoints in this study: mobility fails the cluster-permutation check (FDR p=0.152) despite being well supported under the direct-age sensitivity (S2.4), while the entropy-related coordinate fails **both** the cluster-permutation check (FDR p=0.152) and the direct-age sensitivity (raw/FDR p=0.070) and should be treated as the least settled result of the twelve.

S3.6 Conventional-EEG comparator coverage and robustness (R6)

The conventional-EEG comparator family (S1.8, S2.10-S2.12) uses a second H5 re-extraction, not the primary export used everywhere else in this study (`results/README.md`); every shared MNPS/MNJ/reachability numeric column was confirmed bit-identical between the two exports before use. Its 18 non-connectivity endpoints have 100% session coverage (70/70); its 16 connectivity endpoints have 80–96% coverage, an artifact of per-recording channel/segment availability at specific electrode pairs (F8-P4 and central-parietal/frontal-central pairs lower than F4-P4), not of the analysis code. Applying the identical M3 (within/between), M4 (cluster-permutation), and within-child sign-permutation checks used for the primary family (S3.5) to this 34-endpoint family shows the same qualitative pattern: an asymptotically-supported 24/34 reduces to a permutation-robust 5/34 (a comparable proportional drop to the primary family's 12/12 to 6/12), and the cohort-level effect is overwhelmingly within-child (31/34 FDR-supported) rather than between-child (1/34). The two most extreme cohort-level effects in the **entire study** (alpha frontal-central coherence mean and SD, raw $p \approx 10^{-}64$ to $10^{-}94$) belong to this family and are specifically flagged in the main text as plausibly reflecting an age-correlated non-neural confound rather than a confirmed coherence-maturation effect, given their comparatively lower (80%) session coverage.

Robustness summary

The primary cohort-coordinate age association is attributable to within-child change (not a between-child artifact) and is largely robust to a direct-age sensitivity check (11/12), but only 6 of 12 endpoints survive a conservative, FDR-corrected child-cluster permutation check of the same omnibus test – readers should weight the asymptotic "12/12 FDR-supported" framing accordingly. The within-child and cohort-pooled local MNJ/reachability robustness checks show no FDR-supported change in any tested family; their primary forms are not nominally significant, but the cohort-pooled direct-age subset retains one nominal, uncorrected rotational-proxy hit. A prior pre-masking export also contained two within-child nominal hits and one primary cohort-pooled nominal hit, all without FDR support. All statements remain limited by irregular sampling, modest child count, and a global anchor fitted over all ages. A separate, descriptive 34-endpoint conventional-EEG comparator family (S3.6) shows the same asymptotic-to-permutation fragility pattern as the primary family, and confirms that previously reported qEEG maturation markers (beta/alpha ratio, relative beta/gamma power) are detectable in this cohort and pipeline even where the local-dynamics families (R4, R5) are not.

Bibliography